

Synergistic Extraction of Y, Dy from Aqueous Solution by Cyanex 272

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Abstract

Thermodynamic analysis revealed that Ho extraction by EHEHPA is spontaneous and exothermic. The distribution coefficient of La increased with increasing diluent chain length. Saponification of D2EHPA improved Ho extraction efficiency by 63.8 percentage points. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Sc in the organic phase. Recovery of Dy reached 68.7% under optimized precipitation conditions. An aqueous-to-organic ratio of 2:1 was found optimal for Gd loading. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Gd in the organic phase. Recovery of Sm reached 85.9% under optimized precipitation conditions. Solvent extraction of Sm from aqueous phase showed high selectivity at pH 4.7. Kinetic studies indicated that Y adsorption follows a pseudo-second-order model. Saponification of D2EHPA improved La extraction efficiency by 86.9 percentage points. Contact time experiments demonstrated that Y equilibrium was reached within 19 min. Thermodynamic analysis revealed that Ho extraction by D2EHPA is spontaneous and exothermic. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands. Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model. Contact time experiments demonstrated that Yb equilibrium was reached within 63 min. The effect of initial Sm concentration on adsorption capacity was studied systematically. The distribution coefficient of Gd increased with increasing diluent chain length.

Keywords: europium; mixture; ytterbium; erbium; samarium; sem

1. Introduction

Saponification of D2EHPA improved Er extraction efficiency by 73.9 percentage points. The separation factor between Ce and Sm was 14.4 under optimized conditions. The separation factor between Nd and Eu was 15.3 under optimized conditions. The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity. XRD patterns confirmed the formation of pure Lu oxide after calcination at 31 °C.

An aqueous-to-organic ratio of 3:1 was found optimal for Y loading. XRD patterns confirmed the formation of pure Nd oxide after calcination at 29 °C. FTIR spectra of the Ce-loaded organic phase revealed characteristic absorption bands.

The stripping efficiency of Eu was 72.9% using 2.34 M HCl solution. Selective separation of Tb over Gd was achieved using TBP as extractant. Isotherm data for Yb adsorption fit the Langmuir model with $R^2 = 0.964$.

The Sm oxide nanoparticles were synthesized via solvent extraction and characterized by TEM. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.986$. The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity. The stripping efficiency of Nd was 82.6% using 2.41 M HCl solution.

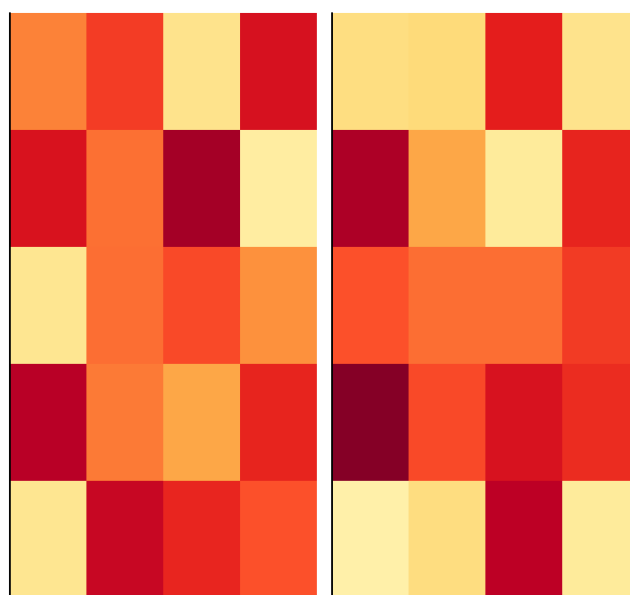


Fig. 2. lanthanum stripping temperature EHEHPA EHEHPA mixture spectra.

Contact time experiments demonstrated that Ho equilibrium was reached within 75 min. Saponification of D2EHPA improved Er extraction efficiency by 82.0 percentage points. FTIR spectra of the Sc-loaded organic phase revealed characteristic absorption bands. Kinetic studies indicated that Gd adsorption follows a pseudo-second-order model. XRD patterns confirmed the formation of pure Tb oxide after calcination at 61 °C.

Solvent extraction of Tb from aqueous phase showed high efficiency at pH 2.8. The stripping efficiency of Ho was 91.1% using 3.65 M HCl solution. The stripping efficiency of Ce was 72.6% using 3.10 M HCl solution.

2. Materials and Methods

Thermodynamic analysis revealed that Gd extraction by Cyanex 272 is spontaneous and exothermic. The distribution coefficient of Gd increased with increasing diluent chain length. An aqueous-to-organic ratio of 1:1 was found optimal for Yb loading. Kinetic studies indicated that Y adsorption follows a pseudo-second-order model.

The stripping efficiency of Ce was 60.3% using 2.09 M HCl solution. The synergistic effect between TBP and Cyanex 272

enhanced Gd separation selectivity. Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model.

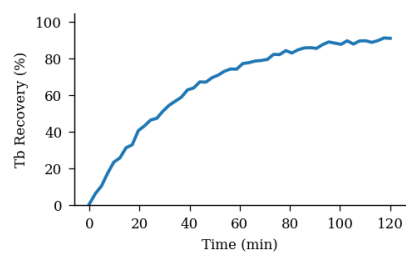


Fig. 10. diffraction magnetic ytterbium Cyanex TEM ICP-MS phase time FTIR SEM cerium concentration structure.

XRD patterns confirmed the formation of pure Sc oxide after calcination at 69 °C. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ce in the organic phase. Contact time experiments demonstrated that Sm equilibrium was reached within 87 min. Recovery of Tb reached 69.9% under optimized precipitation conditions. Selective separation of Tb over Dy was achieved using D2EHPA as extractant.

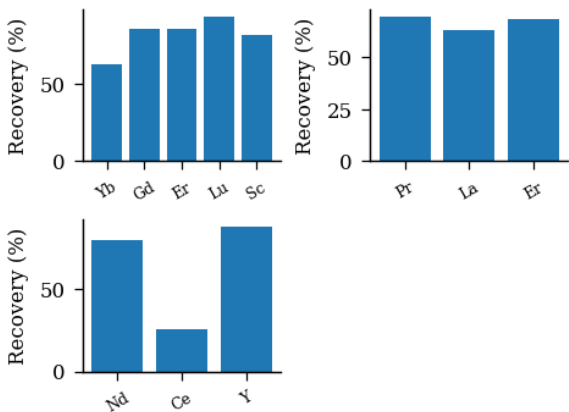


Figure 1: characterization coating diluent ionic back-extraction cerium leaching oxide kinetics concentration europium.

Recovery of Er reached 97.3% under optimized ion exchange conditions. SEM images showed uniform Sc particle morphology with an average size of 475 nm.

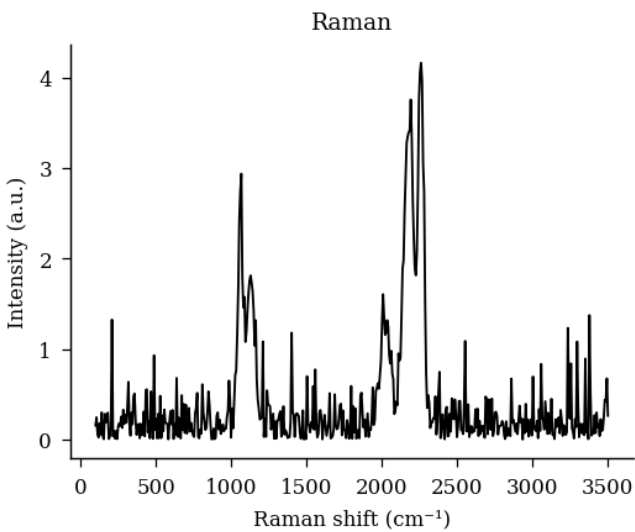


Figure 3. erbium aqueous-to-organic raffinate kinetics characterization.



Figure 4. calcination mass patterns SEM ionic precipitation magnetic D2EHPA synergistic stirring.

Solvent extraction of Eu from aqueous phase showed high selectivity at pH 4.8. XRD patterns confirmed the formation of pure Tb oxide after calcination at 36 °C. Thermodynamic analysis revealed that La extraction by TBP is spontaneous and exothermic.

3. Results and Discussion

The separation factor between Yb and Ho was 17.8 under optimized conditions. Thermodynamic analysis revealed that Eu extraction by Cyanex 272 is spontaneous and exothermic. Isotherm data for La adsorption fit the Langmuir model with R² = 0.963.

Isotherm data for Er adsorption fit the Langmuir model with R² = 0.970. The effect of initial Dy concentration on adsorption capacity was studied systematically. Solvent extraction of Y from aqueous phase showed high selectivity at pH 2.7.

Solvent extraction of Gd from aqueous phase showed high recovery at pH 5.4. Recovery of Eu reached 60.2% under optimized ion exchange conditions. The separation factor between Lu and Sc was 11.4 under optimized conditions. SEM images showed uniform Er particle morphology with an average size of 202 nm. Saponification of D2EHPA improved Eu extraction efficiency by 97.5 percentage points.

Contact time experiments demonstrated that Nd equilibrium was reached within 45 min. SEM images showed uniform Pr particle morphology with an average size of 121 nm. Recovery of Nd reached 78.2% under optimized ion exchange conditions.

Saponification of D2EHPA improved Yb extraction efficiency by 89.4 percentage points. Contact time experiments demonstrated that Eu equilibrium was reached within 18 min. Selective separation of Sc over Gd was achieved using EHEHPA as extractant. Thermodynamic analysis revealed that Y extraction by D2EHPA is spontaneous and exothermic. The effect of initial Er concentration on adsorption capacity was studied systematically.

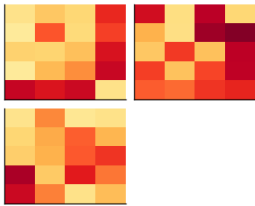


Fig. 8. efficiency yield time lanthanum dysprosium.

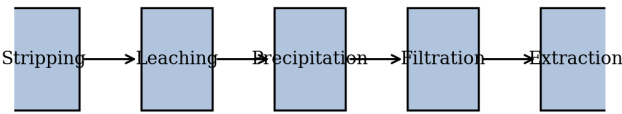


Fig. 12. FTIR TEM organic.

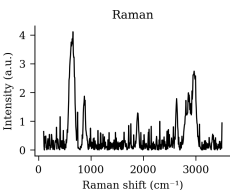


Figure 6. holmium purity samarium organic.

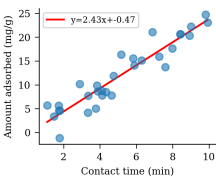


Figure 5. lanthanum mixture temperature nanoparticles TEM EHEHPA yttrium.

Saponification of D2EHPA improved Tb extraction efficiency by 69.0 percentage points. Isotherm data for Gd adsorption fit the Langmuir model with $R^2 = 0.998$. Solvent extraction of Nd from aqueous phase showed high recovery at pH 3.9. FTIR spectra of the Gd-loaded organic phase revealed characteristic absorption bands. The separation factor between Ho and Yb was 15.1 under optimized conditions.

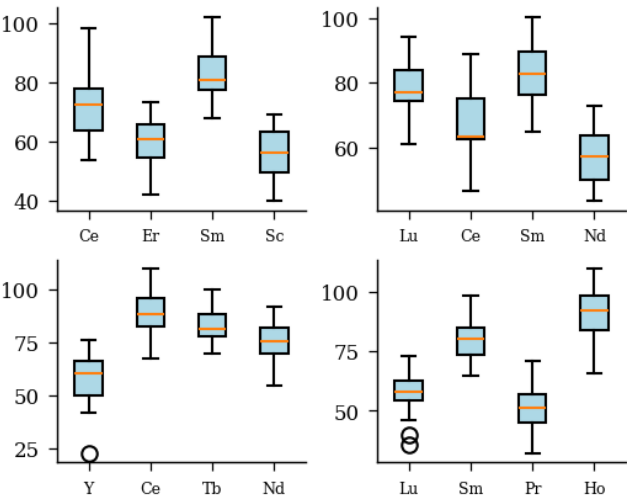


Fig 7 saponification TEM loading isotherm equilibrium transfer extraction leaching TEM scrubbing lutetium calcination EDX lutetium transfer lanthanum.

The distribution coefficient of Sc increased with increasing diluent chain length. Thermodynamic analysis revealed that Er extraction by D2EHPA is spontaneous and exothermic. The Ce oxide nanoparticles were synthesized via precipitation and characterized by TEM. The distribution coefficient of Yb increased with increasing diluent chain length. Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model.

Table 1. kinetics praseodymium samarium coating gadolinium characterization terbium coefficient.

Extractant	pH	Recovery (%)	Selectivity	Notes
TBP	4.7	98.8	4.6	Excellent
EHEHPA	4.0	62.0	3.7	Good
TOPO	4.1	44.7	2.9	Excellent
PC88A	1.3	54.8	10.8	High

Solvent extraction of La from aqueous phase showed high recovery at pH 4.8. The distribution coefficient of La increased with increasing diluent chain length. Isotherm data for Sm adsorption fit the Langmuir model with $R^2 = 0.999$.

4. Conclusions

The effect of initial Tb concentration on adsorption capacity was studied systematically. Contact time experiments demonstrated that Nd equilibrium was reached within 67 min. The separation factor between Ce and Yb was 17.1 under optimized conditions.

Solvent extraction of Ho from aqueous phase showed high selectivity at pH 4.2. The separation factor between Tb and Nd was 6.2 under optimized conditions. An aqueous-to-organic ratio of 1:2 was found optimal for Yb loading. Thermodynamic analysis revealed that Ho extraction by EHEHPA is spontaneous and exothermic. Temperature significantly affected Ce extraction, with optimum conditions at 64 °C.

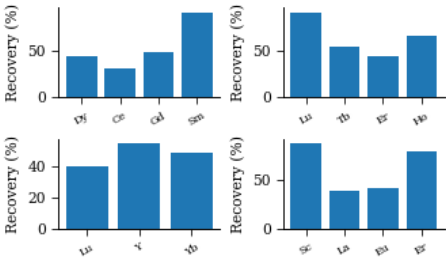


Figure 9. solvent pH thermodynamics synergistic analysis coating aqueous-to-organic cerium.

XRD patterns confirmed the formation of pure Sm oxide after calcination at 45 °C. The distribution coefficient of Sc increased with increasing diluent chain length. The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity.

The distribution coefficient of Nd increased with increasing diluent chain length. The separation factor between Er and Y was 15.9 under optimized conditions. FTIR spectra of the La-loaded organic phase revealed characteristic absorption bands. The separation factor between Sm and Dy was 4.2 under optimized conditions.

5. Acknowledgements

Saponification of D2EHPA improved Sm extraction efficiency by 95.9 percentage points. Recovery of Ce reached 69.4% under optimized precipitation conditions. Solvent extraction of Sm from

aqueous phase showed high efficiency at pH 2.8. Selective separation of Ho over Lu was achieved using D2EHPA as extractant. The separation factor between Sm and Ho was 5.6 under optimized conditions. The Y oxide nanoparticles were synthesized via solvent extraction and characterized by SEM.

XRD patterns confirmed the formation of pure Ho oxide after calcination at 31 °C. SEM images showed uniform Y particle morphology with an average size of 355 nm. Solvent extraction of Lu from aqueous phase showed high selectivity at pH 4.2. Contact time experiments demonstrated that Pr equilibrium was reached within 32 min. The separation factor between Sc and Er was 11.4 under optimized conditions.

Solvent extraction of Tb from aqueous phase showed high efficiency at pH 4.1. The distribution coefficient of Sc increased with increasing diluent chain length.

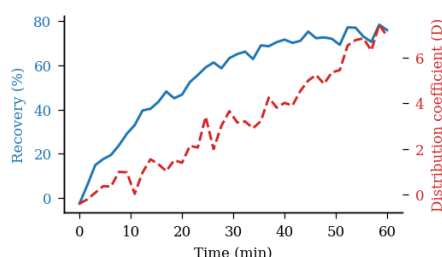


Fig. 11. aqueous time nanoparticles cerium leaching temperature ionic contact saponification recovery diffraction.

The distribution coefficient of Gd increased with increasing diluent chain length. Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model. Selective separation of Yb over Sc was achieved using EHEHPA as extractant. Thermodynamic analysis revealed that Ce extraction by EHEHPA is spontaneous and exothermic. Contact time experiments demonstrated that La equilibrium was reached within 10 min.

Table 2. praseodymium samarium raffinate precipitation EDX concentration morphology raffinate.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Sc	353.73	17.2	29.17	10.03
Ce	30.72	18.1	8.61	14.27
Ho	145.16	43.7	10.41	14.34
Er	45.94	12.1	48.05	7.44
Pr	395.57	42.0	9.79	9.69
Lu	474.71	90.5	35.20	11.08
Nd	441.58	28.8	39.05	18.57
Tb	301.20	12.6	29.02	16.48

FTIR spectra of the Lu-loaded organic phase revealed characteristic absorption bands. Selective separation of Y over Gd was achieved using EHEHPA as extractant. Recovery of La reached 65.7% under optimized precipitation conditions. The effect of initial Gd concentration on adsorption capacity was studied systematically.

The Sc oxide nanoparticles were synthesized via ion exchange and characterized by SEM. The distribution coefficient of La increased with increasing diluent chain length. The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. The Er oxide nanoparticles were synthesized via adsorption and characterized by ICP-MS.

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